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Pharmacology I (Phar 441)
Laboratory Manual

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COMMON LABORATORY ANIMALS

Handling of the laboratory animals involves two most important responsibilities on the part of the experimenter. First, the animal is handled with utmost care so that it does not suffer any pain, and second, a due regard is paid towards the health and well-being of the animal colony. Even when they are killed at the end of the experiment, it should be done by a humane method, i.e. euthanasia, which means painless-killing.

Most of the common laboratory animals such as mice, rats, guinea-pigs and rabbits can be bred and reared under ideal conditions to provide a reliable supply of uniform and healthy animals for the purpose of experimentation. Dogs and cats, however, can be obtained from outside sources as and when required; but it is always better to keep them under quarantine for a few weeks before they are utilized.

Breeding Types

Inbred strains: The primary purpose of inbreeding is to reduce the number of individuals that are heterozygous for any one gene pair, and to increase the number that are homozygous for one or other member of the gene pair. *Coefficient of inbreeding* is the degree of inbreeding as measured by the reduction in the number of heterozygotes, and increase in the number of homozygotes. A strain is regarded as inbred when it has been mated full brothers and sisters (*b x s* or *full-sib mating*) for 20 or more consecutive generations. By this method, most rapid and a maximum increase in the *coefficient of inbreeding* can be achieved. Even then it is not completely inbred in the true sense because it gives an inbreeding coefficient of only 98.6 per cent, which means that on the average 1.4 per cent of the originally heterozygous genes are still heterozygous. Parent and off-spring matings may be substituted for *b x s* matings provided that in the case of consecutive parent x offspring matings, the mating in each case is to younger of the two parents. The laboratory animals most commonly used in their inbred states are mice and rats. The chief drawbacks of inbred strains are that the animals become generally less healthy and more susceptible to disease and their reproductive capacity is reduced (*inbreeding depression*). In experiments on inbred strains the environment need to be more closely controlled than in tests using hybrids or random-bred animals.

F_1 hybrids: The first generation (F_1) of a cross between two highly inbred strains consists of animals that are genetically uniform but are but are not inbred. The F_1 animals unlike inbred strains do not suffer from inbreeding depression and they are improved in viability. Hence, they are less costly to produce than the inbred strains.

Random breeding: It means that the young animals for breeding are chosen without regard to their parentage and mated together without regard to their relationships; the breeding is thus random with respect to pedigree.

Selective breeding: It consists of choosing the individuals to be parents according to the characteristics that it is desired to change. If, for example, it is desired to increase the body weight, then the heaviest individuals are chosen to be the parents of the next generation; or if it is desired to increase the response to a certain drug, then the individuals showing the greatest response are chosen as parents.

20 5
0.5

$$\frac{6}{14} \times 104 = 60$$

Breeding Methods

Hand mating: The male and the female are brought together for a brief period and then separated once when the mating is over. Rabbit and hamster are mated by this method.

Pair mating: One male is mated with one female and left together for the rest of their breeding life in case of mice. In case of rats, pregnant females should be isolated prior to delivery.

Trio mating: One male is mated with two females.

Harem mating: Up to four females are regularly mated to one male. The female is isolated soon after the pregnancy is established and replaced by fresh females. Guinea-pig is mated by this method.

Factors affecting the nature and degree of pharmacological responses

Species: In rabbits intracerebroventricular injection of 5-hydroxytryptamine induced a lowering of temperature, but in cats it induces fever. Ethacrynic acid is almost inactive diuretic in the rat except in extremely high doses, but quite active in the dog. Following the same dose of hexobarbitone per unit body weight, the average sleeping time of rats is about seven times that of mice, and in dog its effect lasts for hours. Guinea-pigs and humans are some 500 times more sensitive to histamine than are rats and mice. Histamine powerfully contracts the uterus of guinea-pig while relaxes that of rat. In rodents, it produces strong arteriolar constriction, in cat slight constriction; while in the dog, monkey and man arteriolar dilatation. The rat heart is known to be very resistant to cardiac glycosides. Alloxan produces diabetes mellitus in a number of species but not in guinea-pig. Rabbits generally show symptoms following insulin when the blood sugar level is 45 mg per cent, while dogs show symptoms only when the level is about 18 mg per cent.

Strains: Due to the occurrence of the hydrolyzing enzyme (atropinesterase) in the serum and liver of some genetically determined rabbits, some strains of rabbits are exceedingly resistant to atropine. Sensitivity variations have also been noted between different strains of mice to the action of histamine, insulin and barbiturates.

Sex: Renal lesions following exposure to chloroform have been observed exclusively in male mice but not in female mice. Female rats are more sensitive to certain barbiturates than are males. The male rat has a faster growth rate than the female; therefore, indiscriminate use of both males and females in a rat growth test should be avoided.

Age: In general, neonates tend to be more sensitive than are adults to the same dose of the drug. For example, morphine is about four times more potent in 14-day old rats than in rats that are 28-day old. Similarly, the LD₅₀ of newborn rats to pentobarbitone is about one-third that of adult rats.

Circadian or biological rhythm: When amphetamine is given at one phase of rat circadian system, 77.6 per cent of them survive, whereas at another phase only 6.6 per cent survive. The duration of sleep following pentobarbitone averages 104 minutes when administered at one phase, while it averages only 43 minutes when administered at another phase. The anti-cancer

drug cytosine arabinoside is far more toxic at one phase of the mouse circadian system than it is at others.

Isolation or group-housing. Doses of pentobarbitone or phenobarbitone, which produce full hypnosis in isolated mice, produce marked stimulation when the animals are grouped. The cardiovascular system becomes more responsive to isoprenaline in solitary-housed (6 to 8 weeks) rats compared to group-housed rats (Speiser and Weinstock, 1974).

Season: The antidiuretic action of vasopressin in rats was found to be greater in spring and summer, and the variability was least in winter. Mice are more resistant to insulin-induced convulsions in summer, and the slope of the regression lines is steeper in winter than in summer. Frogs have been found to be less susceptible to the cardiotoxic effects of ouabain, strophanthin and digitalis during the period of July-October than in the rest of the Year. This is due to the fact that they hibernate, and their metabolic rates are accelerated in the summer months. The stimulation of the vagus never in frogs produces predominantly inhibitory response in winter and acceleration in summer. In rabbits a low incidence of gonadotrophin-induced ovulation occurs in the months of August-October. After estrogen treatment no ovulation occurs in summer and autumn, whilst large numbers ovulate in winter and spring. The rabbit uterus is also more sensitive to estrogen in spring than in autumn or early winter.

THE RAT (*Rattus norvegicus*)

Albino rat is one of the commonest laboratory animals suitable for experimental work because of its small size and greater sensitivity to most drugs. It is also the most standardized of all laboratory animals. It can be bred to obtain pure and uniform strains and is found to be very sturdy to withstand long periods of experimentation under anesthesia. There are two main strains of albino rats that have been widely used throughout the world.

Wistar rat is a quiet, moderately prolific strain resistant to infection, and has a low incidence of spontaneous tumors. The head is wide, especially in the male, and the ears are long. The tail length is always less than the body-length.

Sprague-Dawley rat is more rapidly growing and more prolific than Wistar, has a longer and narrower head, and a longer tail, which may equal the body-length. It is also less resistant to infection, specially to respiratory disease.

There are certain anatomical peculiarities, which should be noted. Rodents do not vomit because they lack vomiting center. Rats do not have any tonsil or a gall bladder. They have extremely diffuse pancreas and hence pancreatectomy is difficult to perform in order to produce a diabetic rat. The stomach shows an obvious division into two parts by a prominent white somewhat curved transverse ridge (limiting ridge). The upper two-fifth glandular nonsecretory portion (rumen) is translucent and thinner than the lower three-fifth glandular secretory portion. Histologically also the two areas show distinct differences. The glandular portion is composed of two clearly defined areas that can be distinguished both by naked eyes as well as under a microscope. The antrum is a roughly semicircular is straddling the lesser curvature below the limiting ridge. It is relatively thin and translucent, and has a slightly irregular inner lining. The



5

Wt = 30 gm
St = 10 mg/ml
D = 80 mg/kg
Calculate dose
10 mg X 1000 gm
30 gm
300 / 1000 = 0.3 mg
20 mg X 1 ml
0.4 mg
0.4 / 10 = 0.04

remainder of the lower two-third comprises the body of the stomach, which straddles the entire greater curvature below the ridge except for a small area above the pylorus. This is the thick portion of the stomach with mucosa thrown into folds. Shay and co-workers (1945) believe that this (body) part of the stomach is analogous to the functionally. The rat being omnivorous resembles man nutritionally.

Oestrus cycle: The cycle makes its appearance at puberty at the age of two to three months, and the whole cycle lasts for about four to five days being divided into four stages according to the cell types found in the vaginal smear. These are as follows:

- i) Oestrus (9 to 15 h) - is characterized by sexual receptivity when the female will allow copulation. During this period there are increased running activity, quivering of the ears, and lordosis in the presence of another rat. The vaginal smear shows 100 per cent cornified epithelial cells.
- ii) Met-oestrus (about 20 h) - follows oestrus and occurs shortly ovulation. The vaginal smear is characterized by many leucocytes with a few cornified cells.
- iii) Dioestrus (60 to 70 h) - is the longest of phases, and vaginal smear consists mainly of leucocytes.
- iv). Pro-oestrus (about 12 h) - which follows dioestrus is a preparatory phase preliminary to the next oestrus phase. The vaginal smear is characterized by nucleated epithelial cells either singly or in sheets.

Vaginal smears may be taken in the following way to confirm the actual stage. Cotton swab made with tooth-picks is moistened with saline and gently inserted and slightly rotated within the vagina. The swab is then pressed in a drop of saline on a microscope slide and examined under low power. An alternative and better method is to take the vaginal wash with a few drops of saline with the help of a capillary pipette fitted with a rubber teat.

Rat as an Experimental Animal

Rats are particularly suitable for testing of psychopharmacological agents because they can be trained properly for various types of work performances including development of conditioned reflexes.

They are also utilized for the assay of different hormones and for study of the oestrus cycle, mating behavior and lactation.

These animals are also employed for the study of drugs on the blood pressure both in acute as well as in chronic or surviving experiments.

The status of the rat as an experimental animal in the study of gastric secretion is unique since a continuous secretion though of a low order appears in the fasting animals in contradistinction to large animals like cat, dog, etc. The mechanism of this true spontaneous secretion is not clear. Since there is some evidence that bile in the intestine is a fairly strong gastric secretagogue, the absence of gall bladder in the rat resulting in a continuous entrance of bile into the intestine may serve as a partial explanation. For the study of gastric secretion, animals are always fasted (water allowed) for a period varying from 24 to 72 hours. Special care should be taken so that they do not get access to their stools, because they are in the habit of eating their own stools (coprophagy). Shay and co-workers in 1945 produced for the first time an acute ulcerative lesion in the fore stomach (rumen) by pyloric ligation. The effects were so uniform that the shay rat

method was subsequently adopted by a number of workers for evaluation of antiulcer drugs. A quantitative method for the assay of inhibitors of acid gastric secretion has also been developed.

Rats are also employed for the study of analgesic drugs by the method of radiant heat application on the tail.

They have been employed routinely in the toxicity studies, both acute and chronic—especially the latter—because the drug can be administered and effects observed throughout their short life span of about three years.

The 24-hours old rat is said to be physiologically similar to a 6-month old infant, and is used for evaluation of compounds supposed to be administered in infants of this age group.

It is particularly suitable for the study of the physiology of the liver since after partial hepatectomy (60 to 70 per cent liver tissue removed) the organ regenerates almost completely in a course of a week.

Rats are suitable for the testing of drugs for teratogenicity and carcinogenicity. Various isolated tissues such as uterus, stomach, colon, etc. of rat are also routinely employed for the study of drug actions. The uterus of the rat is inhibited by adrenaline; this forms a basis of one of the bioassay methods of adrenaline.

THE GUINEA-PIG (*Cavia porcellus*)

The name guinea-pig has become almost synonymous with experimental animal although in fact mice and rats are used in far greater numbers. Guinea-pig or cavy has proved to be the most useful laboratory animal. Like other laboratory rodents, it is easy to keep them in captivity. But being docile in nature, and very susceptible to tuberculosis and to anaphylactic shock, they differ from other laboratory rodents. They are also highly sensitive to histamine. Following histamine or anaphylactic reaction, there is bronchoconstriction, and the animal usually dies from asphyxia. Penicillin is 100 to 1000 times more toxic to the guinea-pig than to the mouse. In this species dopamine causes a fall in blood pressure. The uterus inhibited by adrenaline.

Guinea-pig serum contains an enzyme asparaginase, which has been shown to have some anti-leukaemic action.

Guinea-pig as an Experimental Animal

Guinea-pig is employed for the evaluation of bronchodilator compounds against experimentally induced asthma (histamine or acetylcholine aerosols).

They are widely used in immunology, particularly in the studies of delayed hypersensitivity. Sensitization studies are best conducted in guinea-pigs employing different antigens like egg albumin, horse serum, etc.

They are employed in the study of local anesthetics as well as for the bioassay of digitalis. Guinea-pigs having sensitive cochlea are suitable for hearing experiments.

Being docile in nature and being more resistant to hypoxia than rats and mice, they are suitable for experiments on oxygen consumption.

Being the only common laboratory animal, which resembles man in that it requires exogenous vitamin C, guinea pig is used in the study of ascorbic acid metabolism.

It is a suitable host for mycobacterial infection, and is used in the study of tuberculosis. Guinea-pig seems to be closer humans than rats for the study of isoniazid toxicity.

Various isolated tissues such as ileum, tracheal chain, vas deferens, etc. are utilized for pharmacological studies of different compounds. The terminal ileum is the most common and standard preparation for preliminary screening of spasmodic or antispasmodic compounds, and particularly sensitive to and suitable for the detection and assay of histamine and related compounds.

THE MOUSE (*Mus musculus*)

Albino mice are the smallest laboratory animals, which can be bred uniformly, are cheap and easy to handle, and being very small are sensitive to small doses of a substance. Swiss albino mouse is the most widely used for laboratory investigations.

Mouse as an Experimental Animal

Mice are employed widely in acute toxicity studies. They are also used for the assay of insulin and analgesics, the latter specially against chemically induced pain.

General screening of chemotherapeutic agents is undertaken in this species. Specially bred mice are used for the study of problems in genetics and in cancer research.

Mice are most frequently used for the testing of drugs for teratogenicity. The development of *nude mouse*, a hairless genetic mutant that lacks a thymus and is thus deficient in T lymphocytes, has been of fundamental interest in the field of tissue immunity and transplantation research. They are less susceptible to cancer, the reason being that they possess normal numbers of natural killer cells, which kill both tumor cells and those infected by viruses. Biege mice, on the other hand, which lack killer cells, are unusually susceptible to cancer.

Being very small and delicate the isolated tissues of mice are rarely used excepting vas deferens and ileum.

THE HAMSTER

The species of hamsters are commonly used as laboratory animals: the Syrian or Golden hamster (*Mesocricetus auratus*) and the Chinese hamster (*Cricetulus griseus*).

Hamsters have chunky body with short legs, a diminutive fluffy tail and a large amount of loose skin covered with dense short soft fur. There are four toes on the front foot and five on the back, and the foot pads are hairless. They have prominent cheek pouches extending up to the shoulder region. The ears are quite prominent, rounded, usually pigmented and scarcely covered with hair.

Hamster as an Experimental Animal

Golden hamsters are used for investigation in a number of fields such as virology, cancer and nutrition research, genetics, pharmacology, toxicology and reproductive physiology.

Chinese hamsters are most commonly used in research on diabetes because of the high incidence of spontaneous diabetes mellitus in this species presumably due to the deficiency of B-cells or presence of defective B-cells in the pancreas.

This species is also used as a host in certain parasitological investigations. Chinese hamsters have a low chromosome number (22) compared to other laboratory animals and this makes them excellent tools for cytological investigations, genetics, tissue culture and radiation research.

The presence of cheek pouch in hamsters makes it a very useful animal model for research in the field of immunology. The pouch can be pulled out under anesthesia and can be stretched over an illuminated surface for observation. The pouch in the everted position can be used as a site for the implantation of auto, homo, or heterografts. The cheek pouch of golden hamster is capable of accepting many tumor grafts from other species and their take is nearly 100 per cent.

Strips cut from the other pouch of the golden hamster stomach have been used as useful in vitro test preparation for the assay of prostaglandins E % F.

THE RABBIT (*Oryctolagus cuniculus*)

Rabbits are very docile animals and employed for a variety of studies. New Zealand white rabbits are widely used for the purpose. Genetically the most interesting finding is the linkage between the gene for atropinesterase and that, which influences the colour of the fur (extension of black pigment in the coat). The occurrence of atropinesterase greatly shortens the pharmacologic response to atropine and this can be detected by intravenous injection of 0.1 mg/kg atropine sulfite. In atropinesterase-containing rabbits the pupillary light reflex is either not abolished or it recovers within 10 to 15 minutes, whilst in atropinesterase-free animals it is abolished. The rabbit has got a huge caecum. The afferent cardio-aortic nerve, which in most species including man ascends in the trunk of the vagus, forms a separate nerve bundle called the depressor nerve in the rabbit. The rabbit does not appear to possess adrenergic vasodilator nerves; hence no vasomotor reversal phenomenon is demonstrable in this species. Dopamine causes fall in blood pressure in this species. Histamine tends to raise the blood pressure and produced constriction of the pulmonary vessels. Ovulation in the rabbit is related to the release of luteinizing hormone and occurs about 10 hours after coitus. Such coitus-induced ovulation is inhibited by progesterone. Uterus of the rabbit is stimulated by adrenaline.

Rabbit as an Experimental Animal

Rabbits are employed for the pyrogen testing in intravenous fluids. Agents affecting capillary permeability are studied intercutaneous injection of these compounds followed by intravenous injection of some dyes like Evans blue. Insulin and other antidiabetic drugs, curare and sex hormones are tested in the rabbit. Its skin is relatively sensitive to irritants and so a great deal of experimentation by the dermal routes is being conducted on the rabbit. The convenience of injecting into and withdrawing blood from the ear veins is unique in this laboratory species. Rabbit is routinely used in serological work and for screening embryotoxic agents and teratogens.

It is specially suitable for research on reproduction since ovulation is non-spontaneous, there is no seasonal anoestrus, gestation is short, and semen is easily collected. Isolated heart, duodenum and ileum are some of the preparations routinely employed for the testing of drugs.

THE CAT

Cats are common among the carnivores that are relatively easy to obtain and to use for the experimental purpose. They are used in studies on the physiology of the circulatory and neuromuscular systems because the behavior of these systems corresponds more closely to that of man. The cat has a highly developed nictitating membrane (third eye-lid), which is contracted by stimulation of the cervical sympathetic trunk and also by a number of drugs like adrenaline histamine, etc. In cat, morphine produces excitation of the central nervous system. Histamine produces constriction of the pulmonary vessels and relaxation of tracheal muscles. The depressor nerve in this species may run with the vagus on the right side but separately on the left. Cats are used when stable blood pressure is required. Their vein-walls are much stronger and hence easier to manipulate than those of the pregnant cat is stimulated and that of virgin cat is inhibited by adrenaline.

Cat as an Experimental Animal

Cats are employed in acute experiments for the study of drugs affecting blood pressure. Both anaesthetized and spinal preparations are used latter being particularly suitable for the assay of catecholamines.

Contractions of nictitating membrane are recorded for the investigation of ganglionic blocking action of drugs.

Cats are essential in the study of the nerve centers in the brain.

Because of their ability to produce methaemoglobiemia cats have been found to be most suitable for the toxicity studies of compounds like acetanilide.

THE DOG

Dogs (Mongrel, Beagles, etc.) are useful among the large laboratory animals because they can be tamed as well as trained without much difficulty. Because of a small stomach and short intestinal tract resembling those of man, and because a dog can be conditioned to carrying a stomach cannula, it is the animal of choice for studies in digestion and gastric secretion. Classical work of Pavlov on conditioned reflexes in the dog associating hunger and various signals is well known. In this species the cervical sympathetic and vagal nerves run together inseparably in the same trunk (vagosympathetic nerve) so that the stimulation of nictitating membrane through the preganglionic sympathetic trunk is complicated by central vagal stimulation causing reflex variations in blood pressure.

Dog as an Experimental Animal

Anaesthetized dogs are routinely used in the study of drugs on blood pressure in acute experiments. The size and accessibility of its vascular system coupled with its ability to learn makes the dog invaluable for experiments on circulation that need unanaesthetized animals.

discrete pancreas allowing ready resection enabled the classical work on diabetes to be done in the dog. Since dog develops spontaneous diabetes mellitus resembling man, hypoglycemic drugs can be studied in these animals with advantage.

Chronically prepared gastric fistula and pouches by earlier operations are also employed for the study of gastric secretion in the dog.

THE MONKEY

Monkeys and apes belong to primates, the highest order of mammals which includes man. Both structurally and functionally monkeys and apes closely resemble man. For instance, they are unique among all subhuman mammals in possessing a human type of uterus and in exhibiting regular menstrual periods. The structure of their brain also resembles that of man very closely. The high neurological development of monkeys and anthropoid apes makes them particularly suitable for studies of psychopharmacological agents. Metabolism in rhesus monkey most nearly approximates that of man, and its sensitivity to drugs is also close to that of man.

Monkey as an Experimental Animal

Tests in primates must constitute the last stage in the evaluation of drugs before clinical trials are made.

Primates are used in the fields of virology, parasitology, immunology and immunosuppression, nutrition, reproduction, etc.

THE FROG AND THE TOAD

Since all amphibians absorb moisture through their skin, they must always be provided with water. It is almost impossible to breed frogs and toads in the laboratory without inducement by some hormone. The vagus nerve in the frog is a mixed one containing also the sympathetic accelerator component, that accelerates. The response to its stimulation, therefore, varies from frog to frog and also from season to season. Sometimes an inhibition is produced and sometimes an acceleration; the former predominated in winter, while the latter in summer. Unlike in mammals where nor adrenaline serves as the neurotransmitter, in the frog and toad adrenaline is the transmitter in sympathetic system. Hence, the frog heart falls into the range of the predominate B-adrenergic organs, and it is more sensitive to adrenaline.

Frog and Toad as Experimental Animals

They are commonly used in the study of action of drugs on the central nervous systems, on the heart, on the neuromuscular junction as well as for the diagnosis of pregnancy. For the latter, the "male frog test" is much more rapid than the older tests and equally accurate as well as simple.

Frogs have been used for determining the retinal toxicity of drugs. Light bleaches Rhodopsin from frog retina in one hour and same is resynthesized in one hour in the dark. Certain drugs can be assayed for their toxicity since they prevent the resynthesis of rhodopsin in proportion to the dose given.

Practical No. 1:

Introduction to the Pharmacology Laboratory & Working with Animals

- Introduction to the Pharmacology lab; safety in the lab.
- Animal handling (mice & rabbits)
- Working with animals (weighing, drug administration, anaesthetizing, dissecting and cleaning up when you have finished your work)

Experiment 1: Routes of Drug Administration to Laboratory Animals

Students will work in groups. Each group will be given a mouse on which the following operations will be performed

- weigh the animal & record its weight
- administer 0.5 ml of tap water per os (p.o.)
- administer 0.2 ml of normal saline i.m.
- " " " " s.c. (subcutaneous)
- " " " " i.p. (intraperitoneal)
- " " " " i.v. (into the tail vein)
- Anesthetize the animal with ether (Demonstration by the instructor only)
- Anesthetize the mouse with Na thiopental (1mg/20g) and observe the animal's status (loss of reflex, pattern of respiration, and ease of handling). Write down your observations.

Practical No. 2

General Pharmacology

Experiment 2.1: Influence of Routes of Drug Administration on the Onset and Duration of Drug Action

Objective: To study the onset and duration of action of phenobarbitone sodium in mice when given by different routes of administration

Principle: The onset and duration of action of a drug depends on the route of administration. For example, when a drug is given by intravenous route the effect is seen instantaneously as compared to oral administration of drug when it takes up to one hour to show the effect. Before performing this experiment the students are advised to have background of various routes of administration and their relative merits and demerits.

Phenobarbitone sodium is chosen for the experiment because it shows a clear onset of action of action (animals go to sleep by losing their righting reflex i.e. ability to maintain upright posture) and end point is also clear when the animals recover from sleep. Moreover, its effects can be seen when administered by all routes of administration. For the present purpose, three routes are chosen i.e. intraperitoneal, subcutaneous and oral route of administration of drug.

Requirements

Animal- Mice (25-30 g)

Drugs- Phenobarbitone sodium (Dose: 80 mg/kg. Stock solution contains 10 mg/ml)

Equipment- 1 ml syringe having 100 divisions per ml, hypodermic (26 no.) and oral feeding needles.

Procedure:

1. Weigh the animals and number them by marking them on their tail.
2. Divide them into three groups each consisting of 3 mice, for intraperitoneal, subcutaneous and oral routes of administration.
3. Intraperitoneal (i.p) injection is made in the abdomen whereas the subcutaneous (s.c) injection is made in groin. For oral route, hold the mouse by its neck muscle and place the feeding needle on the tongue and gently push the solution in the mouth. Observe that no drug comes out of mouth or nose. If the solution is coming out from the nose it is sure that the solution is not going in stomach but entering trachea, if not stopped immediately, the mouse may die.
4. Note the time of injection and time of onset of sleep (loss of righting reflex). When the animal sleeps, put it on its back so that when it regains consciousness it will turn over to its normal posture (end point). Animals should be placed apart so that no mouse disturbs the other when it recovers from barbiturate-induced sleep.
5. Note the time of recovery and calculate the duration of action.
6. Compare the onset of action and duration of sleep due to phenobarbitone sodium when given by different routes of administration
7. Observation

No.	Body wt.	Route of admn.	Time of injection (Ti)	Time when asleep (Ts)	Time when wake (Tw)	Onset of action (sec) (Ts-Ti)	Duration of action (min) Tw-Ts)	Mean duration (min)
1.		i.p —		26.57	3.0	—3		
2.		i.p						
3.		i.p						
1.		s.c —						
2.		s.c		26.57				
3.		s.c						
1.		Oral —						
2.		Oral						
3.		Oral						

Conclusion:

Experiment 2.2: Relationship of drug solubility to drug action

Experiment Materials: 4 mice, fasted overnight (2 controls and 2 test animals); 10 ml 30% BaSO_4 soln; 10 ml BaCl_2

Procedures: 1 ml 30% BaSO_4 soln is administered to two mice, while the other two mice receive 1 ml 30% BaCl_2 soln. Both drugs should be administered orally by gavage.

Observe and record the effect produced by the two salts of barium. (Do you observe any behavioural changes between the two groups of animals?)

Conclusions: Explain the results of the experiment.

Experiment 2.3: Dependence of drug action upon liver function

Materials: Four mice (two controls and two test) Carbon tetrachloride (CCl_4) 5% soln; 0.5 % soln of PhB (see Exp. No. 1); syringes and needles.

N.B. Mice in the "test group" have been given 1 ml 5% CCl_4 soln in liquid paraffin for 2 consecutive days before the start of the experiment, while the "control group" was given an equal volume of pure water.

Procedures: Administer to all animals 80 mg/kg PhB used in Exp. No. 1 i.p. and start the stopwatch.

Measure the onset and duration of barbiturate narcosis or sedation in both groups.

Conclusions: Explain the results of the experiment.

Practical No.3

General Pharmacology (Review questions)

Students are required to understand the following basic concepts:

- Receptors
- Drug-receptor interactions
- + - Langmuir equation
- Linear and log dose-response curves
- Affinity and intrinsic activity; pD_2
- Drug antagonism; competitive and non-competitive antagonism; partial agonist
- Schild equation and Schild plot; pA_2
- ED_{50} , LD_{50} , TI, certain safety factor (CSF), standard safety margin (SSM)
- Probit plot

REVIEW QUESTIONS

1. Which one of the following is a true statement concerning drugs classified as noncompetitive antagonists?
 - A. They increase the maximal response to weak agonists
 - B. They generally bind to the receptor through covalent bonds
 - C. Dissociation constants of the antagonist for the receptor are 1000 times higher than dissociation constants for the agonists
 - D. "Spare" receptors are unaffected by these agents
 - * E. They bind to the receptor at different sites than the agonist binding site

Dose-response curves for the therapeutic effect and the toxic effect of Drug A (left panel) and Drug B (right panel) are given in the figure. The responses are plotted as probits.
2. Which one of the following is a true statement concerning the dose-response curves shown above?
 - A. The relative efficacy of Drug A is less than that of Drug B
 - B. The ED_{50} for Drug A is greater than that for Drug B
 - * C. The slopes of the dose-response curves would suggest that Drug A and Drug B are acting at the same receptor to produce their therapeutic effect
 - D. The slopes of the curves for the therapeutic effect and the toxic effect of Drug B would suggest that the two responses occur by the same mechanisms
 - E. Drug A is less potent than Drug B
3. Which drug has the largest certain safety factor?

A. Drug A

x B. Drug B

gap b/w therapeutic dose & toxic dose

- C. Drug A and B are equally "safe"
D. Indeterminate

4. Which one of the following statements is the most correct one concerning these dose-response curves?

- A. If the Y axis were the response plotted as a percentage of maximum rather than probits, the resulting curve would be a hyperbola.
B. If the X axis for Drug B were plotted arithmetically rather than as the log dose, the curves for the therapeutic effect and the toxic effect would be parallel.
C. As both drugs appear to produce the same maximal response, this is clear evidence that they are occupying equivalent numbers of receptors when doing so.
D. As the curves for the therapeutic effect and the toxic effect for Drug B are not parallel, this is evidence that Drug B is a weak agonist.
E. When responses to a drug are converted from the percentage of maximum to probits, equal spacing occurs for each standard deviation.

5. Which one of the following statements is applicable to absorption of drugs from the gastrointestinal tract?

- A. Absorption of weak acids occurs only from the stomach and not from the small intestine.
B. All drugs are stable at the low p^H of the stomach.
C. Rectal administration is the best way to give irritating drugs.
D. Some drugs are metabolized extensively by the liver and do not reach the general circulation (first pass effect).
E. Adsorption to food always enhances drug absorption.

6. Which one of the following statements is true concerning the above dose-response curves?

- A. Curves A, B, and C represent responses to weak agonists with C being the least potent.
B. Curve A represents the responses to a full agonist and curves B and C represent the responses to the agonist in the presence of two concentrations of a competitive antagonist.
C. Curves A, B, and C represent 3 drugs which are acting at different receptors as their potencies are different.
D. Curve D could be the response to agonist A in the presence of a competitive inhibitor.
E. Curve D represents an agonist with a high intrinsic efficacy as the dose needed for a given effect is larger than that of agonists A, B, or C.

7. Which one of the following statements is true concerning renal excretion of drugs?

- A. Drug bound to plasma protein is filtered at the glomerulus.
B. All ionized drugs are reabsorbed in the kidney tubule.
C. Urinary p^H has NO effect on excretion of drugs.
D. Binding of highly ionized drugs to plasma protein will increase their excretion rate.
E. The concentration of drug in the urine is equal to that in the plasma.

8. All of the following statements are true concerning the diminished response of an organ following continued application of drug EXCEPT:

- A. Might result from the modification of a receptor so that it does not bind agonists.
B. Is called tachyphylaxis when it occurs rapidly.

- C. Might result from the internalization of drug-receptor complexes
 D. Involves only binding to spare receptors
 E. May result from depletion of an endogenous mediator
9. Drugs A, B and C act on the same receptor. All of the following are true statements concerning these dose-response curves EXCEPT:
- A. Drug A is a full agonist
 - B. Drug C is a weak agonist
 - C. At sufficient concentration, Drug C could antagonize the effects of Drug B
 - D. Drug A is more potent than Drug C
 - E. Drug B is a full agonist
10. Referring to the same graph in question 9, all of the following statements are applicable to these dose-response curves EXCEPT:
- A. Curve B might be the response to Drug A in the presence of a competitive inhibitor
 - B. Curve C might be the response to Drug A in the presence of a noncompetitive inhibitor
 - C. The intrinsic efficacy of Drug C is less than that of Drug B
 - D. Mixtures of Drug A and Drug B would give a response like that of Drug C
 - E. Drugs A and B are full agonists
11. All of the following are true statements concerning the concept of "spare" receptors EXCEPT:
- A. Maximal effect of partial agonists is obtained only when all receptors are occupied
 - B. Occupation of all receptors by full agonists will produce a maximal effect
 - C. The greater the number of receptors, the greater the possibility of an effect at low concentrations of agonists
 - D. Regular receptors and "spare" receptors have different protein structures
 - E. "Spare" receptors bind to isotope-labeled agonists with the same affinity as do "regular" receptors

Practical No.4

Drug Antagonism

Experiment 4.1: Physiological Drug Antagonism

Materials: - 4 mice (two "control" and two 'test' animals)
 - Syringes with needles
 - 0.5% phenobarbitone sodium (PhB) solution.
 - Stop-watch

Procedure: Administer PhB soln. (Dose = 80mg/kg) to both groups of mice Did the animals become sedated after a few minutes?

Now administer to the 'test' group 1% amphetamine soln. (1ml/kg)

Conclusions: Write down your observations and explain the results you have observed.

Experiment 4.2: Physical Antagonism

Materials: - 4 mice (two "control" & two "test" animals)
- Syringes with needles
- 1% strychnine sulphate/HCl soln.
- Activated charcoal

Procedure: Administer 0.5ml of 1% strychnine sulphate soln. Which has been boiled with activated charcoal to the "control" group orally & give the same dose of strychnine sulphate soln. alone to the "test" group.

Conclusions: Write down your observations & explain the results you have observed.

Practical No.5

Pharmacodynamic Effects of Pilocarpine and Atropine on Rabbit's Eye

Objective: Determination of the diameter of the pupils of the eye of the rabbit and their response to light after administration of pilocarpine and atropine.

Requirements:

Animal- Rabbit (Weight)

Drugs- Atropine solution (1%) and pilocarpine solution (2%)

Equipments- ruler, electric torch, dropper pipette.

Principle: The eye is supplied both by sympathetic and parasympathetic nerves. The superior palpebral muscle and the dilator pupillae (radial fibers) of the iris have sympathetic supply. The sphincter pupillae (circular fibers) of the iris has parasympathetic supply which exercises dominant control. The ciliary muscle is also supplied by the parasympathetic nerves and when it contracts, the ciliary body is moved inwards and forwards. Because of this the lens bulges forward and the eye is accommodated for near vision (spasm of accommodation). The opposite effect is produced by the relaxation of ciliary muscle resulting in paralysis of accommodation (cycloplegia). When the pupil dilates, the iris folds back near the opening of the canal of Schlemm and the drainage of aqueous humor is decreased thereby increasing the intraocular pressure. Constriction of the pupil, by opposite action, will increase drainage and reduce intraocular pressure.

Topically applied drugs can affect the eye by changing conjunctival congestion, papillary size, light reflex, corneal sensitivity and intraocular pressure. However, the effect of drugs on papillary size, light reflex and corneal reflex can be easily studied. The papillary size can be measured by palping a transparent plastic ruler in front of the eye but as close as possible (without touching the eye ball). Light reflex is elicited by directing the light of a torch towards the pupil. The sensitivity of the cornea is tested by gently touching the cornea with a fine cotton swabstick from side and not from the front of the eye. This elicits corneal reflex, which manifests as blinking of the eyelids.

Procedure:

1. Place the rabbit in the rabbit box keeping the head outside.
2. Observe the size of the pupil in both the eyes.
3. Examine the effect of light reflex by holding the torch in front of the eye moving the light beam to and fro.
4. Examine the corneal reflex by touching a side of the cornea with a cotton piece.
5. Instill a few drops of atropine solution in the conjunctiva (4-6 times) over a period of 8-10 minutes in the right eye of the rabbit. The left eye would serve as control. Instill normal saline in the left eye.
6. Record the pupillary size, light reflex and corneal reflex after ten minutes of drug instillation and tabulate the results.
7. Repeat the experiment with pilocarpine using a different rabbit.

Observations

Drug	Pupillary size (mm)	Light reflex (present/absent)	Corneal reflex (present/absent)
Saline	1		
Atropine	1.4	X	—
Pilocarpine	0.5	X	—

blinking
→ RE 1.2 cm
→ RC 1

Conclusion:

Practical No. 6
Virtual Cat & Virtual Rat
(Demonstration with simulated computer program)

Practical No. 7

General and Local Anesthetics

1. General Anesthetic Agents

Experiment 1.1: Comparative anesthetic potency of ether and halothane.

0.5ml, 1ml, 1.5ml and 2ml of each ether and halothane is placed in Desiccators and a mouse placed in each one. Onset, intensity and duration of anesthesia are noted for each anesthetic agent.

Tabulate your results using the parameters monitored and assigning the following scale:

	++++	+++	++	+	0
Onset	Very fast	Fast	Moderate	Slow	No effect
Intensity	Very deep	Deep	Moderate	Shallow	No effect
Duration	>30min.	20-30	10-20	1-10	No effect

2. Local Anesthetics

Experiment 2.1: Action of a local anesthetic on the corneal sensitivity of Rabbit

Objective: determination of the abolition of the blink reflex by a 1% solution of tetracaine (amethocaine).

- Materials:
- rabbit
 - 1% tetracaine (amethocaine)
 - Ophthalmic dropper
 - Fine cotton-wool

Procedure: 3 drops of a 1% solution of tetracaine are instilled onto the surface of the left eye of a rabbit, while 3 drops of distilled water are applied to the right eye. After 10 minutes, each eye (cornea) is touched by a cotton-wool probe 6 times in quick succession. The percentage abolition of the blink reflex is noted and a conclusion drawn.

Practical No. 8

Despair Swim Test in Rats

Objective: To study the antidepressant effect of imipramine in mice.

Principle: Behavioral despair was proposed as a model to test for antidepressant activity by Porsolt *et al.* It was suggested that mice or rats forced to swim in a restricted space from which

they cannot escape are induced to a characteristic behavior of immobility. This behavior reflects a state of despair, which can be reduced by several agents, which are therapeutically effective in human depression.

Requirement:

Animals- Rats (Weight)

Drug- Imipramine (20 mg/kg i.p, stock solution contains 20 mg/ml)

Equipment- 1 ml syringe having 100 divisions per ml with hypodermic (26 no.) needle.

Procedure: Male rats weighing 160-180 g are used. They are brought to the laboratory at least one day before the experiment and are housed separately in cages with free access to food and water. Naïve rats are individually forced to swim inside a vertical Plexiglas cylinder (height: 40 cm, diameter: 18 cm, containing 15 cm of water maintained at 25°C). Rats placed in the cylinders for the first time are initially highly active, vigorously swimming in circles, trying to climb the wall or diving to the bottom. After 2-3 min activity begins to subside and to be interspersed with phases of immobility or floating of increasing length. After 5-6 min immobility reaches a plateau where the rats remain immobile for approximately 80% of the time. After 15 min in the water the rats are removed and allowed to dry in a heated enclosure (32°C) before being returned to their home cages. They are again placed in the cylinder 24 h later and the total duration of immobility is measured during a 5 min test. Floating behavior during this 5 min period has been found to be reproducible in different groups of rats. An animal is judged to be immobile whenever it remains floating passively in the water in a slightly hunched but upright position, its nose just above the surface. Imipramine (20 mg/kg i.p) is administered to the animals and the duration of immobility is determined 1 hr later by placing the animals in the same cage for 5 min.

Observation

S. No.	Body wt.	Dose	Duration of immobility during 5 min (sec)		
			Before	After treatment	% change in activity
1.					
2.					
3.					
1.					
2.					
3.					

Conclusion:

Practical No.9

Psychopharmacologic Agents and Drugs Used in Motor Disorders

Topics for discussion during the practical session:

- Mental illness; the biochemical basis of mental illnesses
- Schizophrenia and its treatment
- Anxiety and its treatment
- Depression and mania and drugs used for their treatment

Classification of psychotherapeutic agents (*neuroleptics or major tranquilizers, anxiolytic sedatives or minor tranquilizers; antidepressants; psychomotor stimulants or psychostimulants; psychodysleptics/hallucinogens/ psychotomimetics*).

Mechanisms of action, major effects, side effects, uses, drug interactions, precautions and contraindications concerning the different classes of psychopharmacological agents

Motor Disorders. *Epilepsy; Parkinson's Disease; Huntington's Chorea* ;
(aetiology; clinical symptoms; drug treatment; adverse drug reactions).

Experiment 9.1: Action of Diazepam on Muscle Tone

Mice (3) are trained to maintain themselves on a rotarod. Mice that cannot stay for 5 minutes on the rotarod are excluded from the experiment. Diazepam (15 mg/kg) is administered i.p. to each mouse and after 20 minutes the rotarod test is repeated. Observations are entered into the lab book and conclusions drawn.

Experiment 9.2: Potentiation of the Effect of Diazepam by Chlorpromazine

The mice used in experiment no. 1 above are individually placed in activity chamber and the number of square crossings is recorded for 5 minutes. Chlorpromazine is then administered (1 mg/100gm) s.c. and the experiment repeated after 5 mins. Conclusions are made.

Practical No. 10

Neuropharmacologic Agents and Drugs Used in Motor Disorders (Cont.)

Experiment 10.1: Anticonvulsant Activity of Phenobarbitone.

Objective: To study the anticonvulsant property of phenobarbitone against pentylenetetrazol-induced clonic convulsions in mice.

Principle: Pentylenetetrazol is a central nervous system stimulant. It produced jerky type of clonic convulsions in rats and mice. The convulsive effect of this drug is considered to be analogous to petit mal type of convulsions in man. Pentylenetetrazol acts through GABA-benzodiazepine receptor mechanisms in brain. It is widely used as a tool in experimental pharmacology to study convulsions and anticonvulsant action of drugs.

Requirements:

Animals- Mice (Weight).

Drugs- Pentylenetetrazol (Dose 80 mg/kg i.p. stock solution contains 8 mg/ml, Phenobarbitone dose 80 mg/kg i.p. stock solution contains 8 mg/ml).

Equipment- 1 ml syringe having 100 divisions per ml with hypodermic (26 no.) needle.

Procedure:

1. Weigh and number the animals. Divide the animals into two groups each comprising of 2 animals. One group is used for studying the effects of pentylenetetrazol alone (control) and the other for studying the protective effect of phenobarbitone.
2. Inject pentylenetetrazol to control animals and note the onset of action (indicated by Straub's tail, jerky movements of the whole body and convulsions) and the severity of convulsions due to drug.
3. Inject phenobarbitone to the second group. After 30 min inject pentylenetetrazol to these animals which have received phenobarbitone. Note onset and severity of convulsions.
4. Note either delay or complete abolition of convulsions in mice treated with phenobarbitone.
5. **Observations**

No.	Body wt.	Treatment	Dose	Convulsions		
				Onset (sec)	Nature & severity	Death/recovery
1.					Jerky movements, Straub's tail, continuous clonic convulsions	
2.						
1.						
2.						

Conclusion:

Neuropharmacologic Agents and Drugs Used in Motor Disorders (Cont.)

Experiment 11.1: Phenothiazine induced catatonia in rats

Greater of catatonia was got in lesser animal no. & more

Objective:

- To study drug (phenothiazine) induced catatonia (extrapyramidal side effect) in rats.
- To study the anti catatonic (anti-parkinsonian) effect of atropine.

Principle: Phenothiazine and butyrophenone type of anti-psychotic drugs are known to produce extra pyramidal side effects in man. These effects, such as akinesia, rigidity and tremors, are called Parkinson's like because in Parkinson's disease the major clinical symptoms include difficulty to move and change posture (akinesia and rigidity) and tremors. These effects of anti-psychotic drugs are due to excessive blockade of dopamine receptors in the extrapyramidal motor system. Therefore, phenothiazines (chlorpromazine or perphenazine) are commonly used to produce Parkinson's like extrapyramidal symptoms in laboratory animals and to study anti-Parkinsonian drugs.

Requirements:

Animals- Rats (Sex:) Weight:)

Drugs- Chlorpromazine- dose 5 mg/kg i.p (Stock Solution 1 mg/ml)

Atropine-dose 2 mg/kg i.p (Stock solution 0.4 mg/ml)

Equipment- Two wooden blocks, one being 3 cm high and the other 9 cm high, 1 ml syringe having 100 divisions per ml with hypodermic (26 no.) needle.

Procedure:

1. **Weigh and number the animals.** Divide the animals into two groups, one for control (for study of chlorpromazine effect) and the other for studying the effects of atropine. Each group should consist of at least six animals.
2. **Inject chlorpromazine** to control animals. Observe severity of catatonia response as follows:
 - Stage I: Rat moves normally when placed on the table, Score = 0
 - Stage II: Rat moves only when touched or pushed, Score = 0.5
 - Stage III: Rat placed on the table with front paws set alternatively on a 3 cm high block, fails to correct the posture in 10 seconds, Score = 0.5 for each paw with a total of 1 for this stage.
 - Stage IV: Rat fails to remove when the front paws are placed alternatively on a 9 cm block, Score = 1 for each paw with a total of 2 for this stage.Thus for a single rat, the maximum possible score would be 3.5 revealing total catatonia.
3. **Observe the severity of catatonia** at 5, 15, 30, 45, 60, 90 and 120 min after chlorpromazine.
4. **To the second group inject atropine** and after 30 min inject chlorpromazine to these animals already treated with atropine. Observe and score the catatonia as in step 2.
5. **Compare the onset and severity of catatonia response** in both the groups. Plot a graph, time along the X axis and catatonic response along the Y-axis. Note the difference in the onset and severity of catatonic response in both the groups.

Observations

Sl No.	Body Wt	Treatment Dose (mg/kg)	Degree of catatonia					
			5	15	30	45	60	90 120

Conclusion:

Experiment 11.2: Effects of Chlorpromazine and Caffeine on locomotor activity
Objective: To study the central nervous system depressant property of chlorpromazine using locomotor activity test.

Principle: Most of the central nervous system acting drugs influence the locomotor activities in man and animals. The CNS depressants such as barbiturates, alcohol and tranquillizers reduce the motor activity while the stimulants such as amphetamine and caffeine increase the activity. In other words, the locomotor activity can be an index of wakefulness (alertness) of mental activity.

Requirements:

- Animals-** Mice (Weight _____, Sex: _____)
- Drugs-** Chlorpromazine -dose 3 mg/kg i.p (Stock solution 0.3 mg/ml)
 Caffeine -dose 8 mg/kg i.p (Stock solution 0.8 mg/ml)
- Equipment-** Activity cage, 1 ml syringe having 100 divisions per ml with hypodermic (26 no.) needle.

Procedure:

1. Weigh the animals and number them.
2. Make squares of equal size in the cage provided to you. Place individually each mouse in the cage for 10 min. note the number of squares crossed by the mouse.
3. Divide the animals into two groups consisting of 3 animals each. Inject chlorpromazine to one group of animals and caffeine to other group and after 30 min re-test each mouse for activity scores for 10 min. note the difference in the activity before and after chlorpromazine.
4. Calculate the percent decrease/increase in motor activity.

Observations

S. No.	Body wt.	Dose	Locomotor activity (scores) in 10 min		
			Before	After treatment	% change in activity

Conclusion:

Practical No. 12

Analgesic Effect of Morphine in Hot Plate Method

Objective: To study the analgesic effect on morphine in mice using hot plate method.

Principle: In this method, heat is used as a source of pain. Animals are individually placed on hot plate maintained at constant temperature (55°C) and the reaction of animals, such as paw licking or jump response is taken as end point. Analgesics increase the reaction time. Eddy and Jambach first described this method and hence this method is also referred as Eddy's hot plate analgesia.

Requirements:

Animals- Mice

Drug- Morphine sulphate (Dose 5 mg/kg , s.c. Stock solution contains 0.5 mg/ml).

Equipments- Eddy's hot plate, 1 ml syringe having 100 divisions per ml with hypodermic (26 gauge) needle.

Procedure:

1. Weigh and number the mice.
2. Take the basal reaction-time by observing hind paw licking or jump response (whichever appears first) in animals when placed on the hot plate maintained at constant temperature (55°C). Normally, animals show such response in 6-8 sec. A cut off period of 15 sec is observed to avoid damage to the paws.
3. Inject morphine to animals and note the reaction time of animals on the plate at 15, 30, 60 and 120 min after the drug administration. As the reaction time increases with morphine, 15 sec is taken as maximum analgesia and the animals are removed from the hot plate to avoid injury to the paws.
4. Calculate percent increase in reaction-time at each interval.

Observation

No	Basal reaction time (sec)	Reaction time (sec) after morphine administration
1.		
2.		
3.		
4.		
5.		

Conclusion